

USER'S MANUAL

Receipt Printer

RP-600



Partner Tech Corporation

Declaration

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Version 1.0

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Warning and Caution



Warning: Items shall be strictly followed to avoid injury or damage to body and devices.



Caution: Items with important information and prompts for operating the printer.

Safety instruction

Before installing and using the printer, please read the following items carefully.

1. Safety warning



Warning: Do not touch the cutter of the printer.



Warning: The print head is a thermal element and it is at a high temperature during printing or just after operation, therefore do not touch it or its peripherals for reasons of safety.



Warning: The print head is an ESD-sensitive device. To avoid damage, do not touch either its printing parts or connecting parts.

2. Caution

- 1) Install the printer on a flat and stable place.
- 2) Reserve adequate space around the printer so that convenient operation and maintenance can be performed.
- 3) Keep the printer away from water source, and avoid direct exposure to sun light, strong light and heat of fire.
- 4) Do not use or store the printer in a place exposed to high temperature, moisture and serious pollution.
- 5) Do not place the printer on a place exposed to vibration or impact.

- 6) No dew condensation is allowed to the printer. In case of such condensation, do not turn on the power until it has completely evaporated.
- 7) Connect the DC adapter to an appropriate grounding outlet. Avoid sharing a single electrical with large power motors and other devices that may cause the fluctuation in voltage.
- 8) Disconnect the DC adapter when the printer is not used for a long time.
- 9) Do not let the water or electric materials (like metal) go into the printer. If this happens, turn off the power immediately.
- 10) Do not allow the printer to start printing when there is no recording paper installed; otherwise the print head and platen roller will be seriously damaged.
- 11) To ensure the print quality and normal lifetime, use recommended or good quality paper.
- 12) Shut down the printer when connecting or disconnecting interfaces connectors to avoid damage to the control board.
- 13) Set the print darkness to a lower grade as long as the print quality is acceptable. This will help to keep the print head durable.
- 14) The printer should only be disassembled or repaired by a technician, who is certified by the manufacturer.
- 15) Keep this manual safe and at hand for ready reference.

Caution: The following content about cutter only applies to printers with cutter.

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1 Introduction

1.1 Outline

RP-600 is developed for high-end thermal receipt printing market, which is widely used.

RP-600 can be connected with other devices via parallel, serial, USB, Ethernet, WLAN. It provides drivers for operating systems such as Windows 2000/XP/Server2003/VISTA/Server2008 /WIN7, Linux, Mac, and UPOS middleware.

1.2 Features

- Low noise, high printing speed
- Easy paper loading
- Easy operation and maintenance
- Support marked paper and continuous paper
- Compatible with various paper widths between 57.5~82.5mm
- Cutting paper automatically
- Cash drawer control interface
- Communication interface optional
- Compatible with ESC/POS commands
- Volume of buzzer adjustable
- Paper saving function
- Low power consumption design

2 Specification

2.1 Technical specification

Item		Parameter
Printing	Print method	Direct thermal line printing
	Resolution	180 × 180 DPI
	Paper width	57.5~82.5mm, continuously adjustable
	Print width	Max. 72mm
	Print speed	Standard: Max.230mm/s Gray scale: Max.150mm/s
	Flash LOGO	Max. 512K bytes
	Data buffer	64K bytes, 4K bytes or 45 bytes
Interface	Fixed USB on board	USB2.0 (full speed)
	Optional interface	Serial (RS-232C) Parallel (IEEE1284) Ethernet WLAN
	Cash drawer	Controlling 1~2 cash drawers can be selected
Memory		RAM: 2MB, Flash: 4MB/2MB
Printer status detection		Paper end sensor/ black mark sensor, paper near end sensor, top cover position sensor, print head temperature sensor, printer voltage detection
Cutter mode		Full cut, partial cut, tear-off
Barcode	1D	UPC-A, UPC-E, JAN8 (EAN8), JAN13 (EAN13), CODE 39, CODE 93, CODE 128, ITF, CODABAR, GS1-128, GS1 DataBar
	2D	PDF417, QR, Maxicode, 2D GS1 DataBar, Composite Symbology
Fonts		Font A: 12 × 24 Font B: 9 × 17 Kanji font A: 24 × 24
Character set		95 Alphanumeric
		14 types of international characters
		128 x 57 code page
		Traditional Chinese, Simplified Chinese (GB2312/18030), Japanese, Korean, English, HK (optional)
		User-defined font (95) & code page
Character enlargement		All characters can be enlarged 1-6 times horizontally and vertically
Character rotation		Rotation printing in four directions (0°, 90°, 180°, 270°)
Command		ESC/POS compatible
Paper	Paper type	Thermal continuous paper, Thermal marked paper
	Paper roll OD	Max. 83mm
	Paper thickness	0.06mm~0.10mm
Power	Input	100-240VAC, 50-60Hz

supply	Output		24V ± 5% DC, average current 2.15A
	External only		Power adapter
Human-machine interface		Power switch	Micro switch
		Button	FEED button
		LED	POWER LED, ERROR LED
		Buzzer	Integrated internal buzzer, the volume can be adjusted
Reliability	Lifetime of print head		≥150Km (standard test sample with 12.5% duty ratio)
	Lifetime of cutter		1,500,000 cuts (under 0.06mm paper thickness)
	MCBF		60,000,000 lines
	MTBF		360,000 hours (main control board)
Operation temperature and humidity			5°C to 45°C, 20% to 90%RH (40°C)
Storage temperature and humidity			-40°C to 60°C, 20% to 93%RH (40°C)
Overall dimensions			196mm×146mm×148mm (L×W×H)
Functions	Saving paper		Support
	Feed button configuration		Support （configure the printer without computer）
	Right-up-side printing		Support
	Water mark printing		Support
	Gray scale printing		Support

2.2 Cutter parameters

Item	Parameter	Remarks
Cutting method	Sliding blade	
Cutting time	500ms	The time of one cut
Cutting interval	3s	20 cuts/min. (Max.)
Applicable paper types	0.06~0.15mm	Thermal paper or normal paper of the same thickness
Operation voltage	24VDC	
Max. operation current	1.5A	24V DC
Cutter lifetime	1500000 cuts (0.06mm paper thickness)	* Including full cut and partial cut

*Full cut: Cut off the paper completely;

Partial cut: Cutting with one point in the middle left uncut.

2.3 Paper specification

2.3.1 Thermal continuous paper

- Paper type: thermal continuous paper, thermal marked paper
- Paper supply method: paper roll
- Paper width: 57.5-82.5mm continuously adjustable
- Paper thickness: 0.06mm-0.10mm
- Thermal sensitive layer: outward
- Paper roll specification: paper roll OD Max ϕ 83 mm
- Paper recommended:

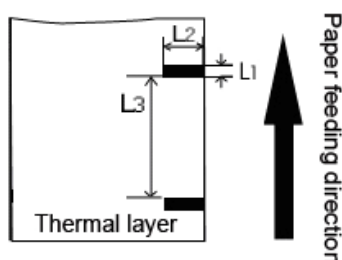
Paper model	Manufacturer
F240AC, F220-VP, FV230A1, PA220AG, HP220A	Mitsubishi Paper Mill CO., LTD
FD210, PD150R, PD160R	OJI Paper CO., LTD
F70NA	FUJI PHOTO FILM CO., LTD

! Caution:

- ✧ Please use the recommended paper or its equivalents. Using other types of paper may affect print quality and print head lifetime.
- ✧ If the paper is contaminated by chemical or oil, it may discolor or lose heat sensitivity at the polluted spot.
- ✧ Do not rub the paper surface strongly against hard objects; otherwise it may make the printing not clear.
- ✧ When the temperature exceeds 70°C, the thermal layer will fade. Thus, do not use or store the paper in a place exposed to high temperature, high humidity, strong light, etc.

2.3.2 Marked paper

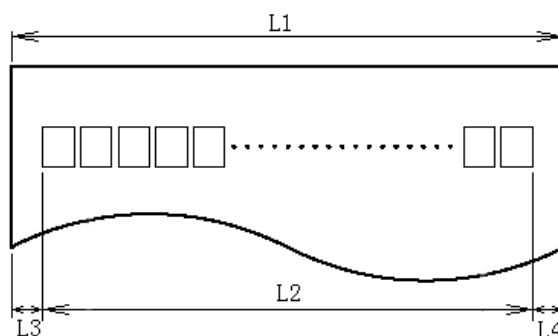
RP-600 can support marked paper printing and set the cutting and initial printing position accurately. Marked paper not only should meet the parameters of thermal paper roll, but also should meet the following requirements:



- L1 mark height: $2\text{mm} \leq L1 \leq 10\text{mm}$
- L2 mark length: $2\text{mm} \leq L2$
- L3 distance between two marks: $20\text{mm} \leq L3 \leq 300\text{mm}$
- Mark position: Any position, default is at the left side
- Reflectivity: The reflectivity of the black mark must be no more than 15% while the reflectivity of the paper itself should exceed 85%.

2.4 Print and tear-off position

2.4.1 Print position



Print position schematic diagram

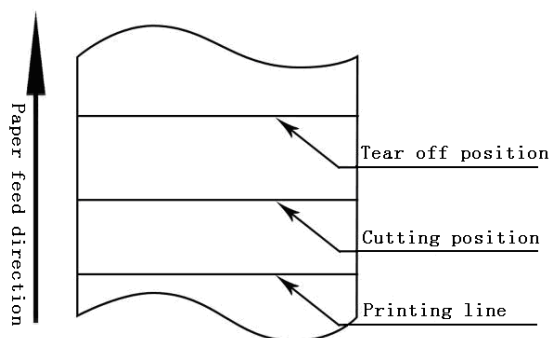
L1 print paper width: 80 mm

L2 valid print width: 72mm

L3 left margin: Default is 4mm

L4 right margin: Default is 4mm

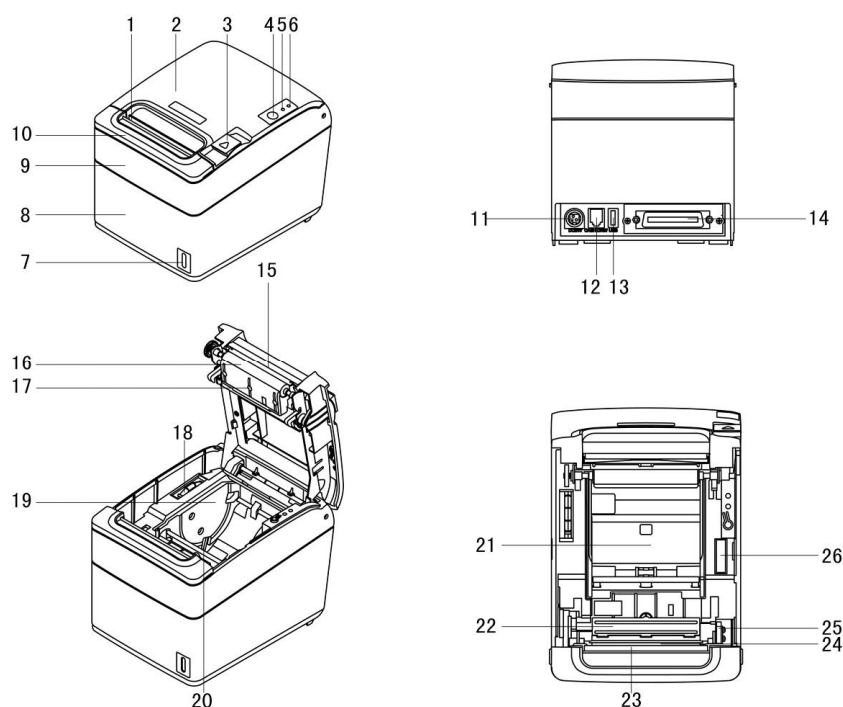
2.4.2 Tear-off position



The distance between the tear-off position and the printing line is 30mm, and the distance between the cutting position and the printing line is 13mm.

3 Appearance and components

3.1 Appearance and components



- | | | |
|--------------------------|--|--------------------------|
| 1—tear-off bar | 2—top cover | 3—top cover lever |
| 4—feed button | 5—error LED | 6—power LED |
| 7—power switch | 8—bottom cover | 9—middle cover |
| 10—cutter cover | 11—power interface | 12—cash drawer interface |
| 13—USB interface | 14—interface board | 15—sliding blade |
| 16—platen roller | 17—upper mark sensor holder | 18—paper near end lever |
| 19—paper guide | 20—paper near end sensor | 21—paper cabinet |
| 22—paper end/mark sensor | 23—fixed blade | 24—print head |
| 25—micro switch | 26—thumb wheel for adjusting paper guide | |

Component function:

a Feed button (4)

➤ Feed paper:

Printer will feed paper when the feed button is pressed down under normal condition. To feed continuously, keep pressing the button.

➤ Print self-test page:

Pressing down the feed button while turning on the power, the printer will print out the self-test page, which includes print length, print width, print speed, etc..

➤ Enter button configuration mode:

Press down the feed button while turning on the power, the printer will print out the self-test page and enter pause state (error LED flashes) after cutting paper. Keep pressing the feed button at this time, the printer will enter button configuration mode.

b Error LED (5)

Indicate printer status. Under normal status, error LED is off. Under error status (e.g. paper end, etc.), error LED flashes.

c Power LED (6)

Indicate printer power status (ON/OFF).

d Power switch (7)

Press down “O” to turn off power, and press down “—”to turn on power.

e Micro switch (25)

Detect top cover status (open/closed).

f Paper end/mark sensor (22)

Under continuous paper mode, it can be used to detect whether paper is present or not; under marked paper mode, it can be used to detect the mark of the paper.

g Paper near end sensor (20)

Detect the paper roll status. Fast flashing of error LED indicates that the paper will be used out soon and the user should replace the paper roll in time. Printer will work normally until paper is used out.

h Thumb wheel for adjusting paper guide (26)

Adjust the thumb wheel for adjusting paper guide to adapt to paper of different widths between 57.5-82.5mm.

3.2 LED and buzzer

1) Functions of LED

LED name	Status	Description
Power LED (green)	Always on	Printer power is on
	Off	Printer power is off
Error LED (red)	Off	Printer is in normal status
	Flash	Printer is in error status or paper near end status
Buzzer	Beep	Printer is in error status

2) Error type indicated by LED & buzzer

Error Type	Error LED	Buzzer
Print head overheating	Cycle flash 6 times	Cycle beep 6 times
Printer voltage abnormal	Cycle flash 5 times	Cycle beep 5 times
Cutter error	Cycle flash 4 times	Cycle beep 4 times
Print head lift-up	Cycle flash 3 times	Cycle beep 3 times
Paper end	Cycle flash 2 times	Cycle beep 2 times
Paper near end	Flash slowly	Not beep
Mark missing or calibration failure	Flash slowly	Not beep

 Caution:

- ✧ The temperature of the print head is detected by a thermal resistor. If the print head is overheating, the protective circuit will shut off the power automatically and force the printer to stop printing; the temperature of print head when printing is stopped is 65°C.

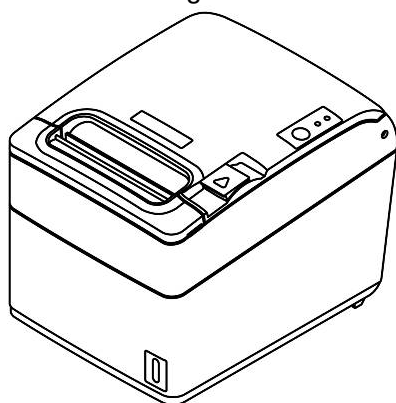
4 Installation

4.1 Unpacking

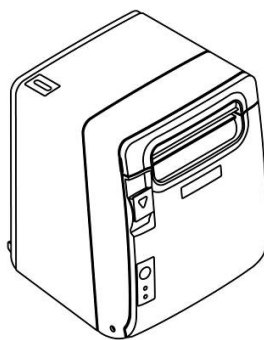
Check whether all items, which are listed on the packing list, are present and in a good condition. If any item is damaged or missing, please contact your dealer or the manufacturer.

4.2 Printer installation

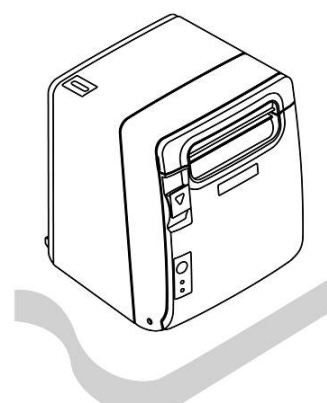
- 1) RP-600 can be installed in three methods: horizontally on the table, vertically on the wall and vertically on the table. When the printer is installed horizontally on the table, the incline installation angle should not exceed $\pm 15^\circ$;



Horizontal installation

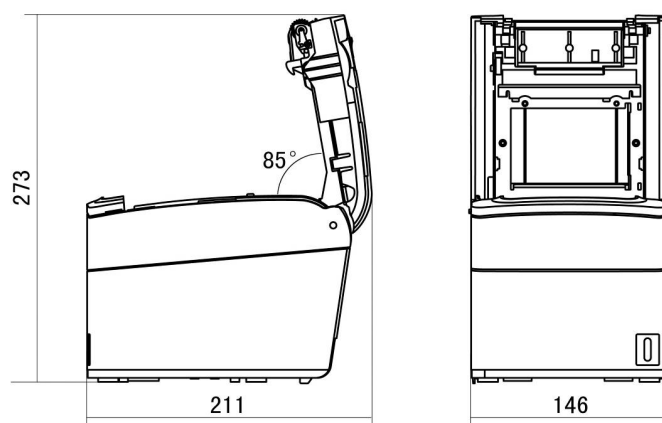


Walling mounting

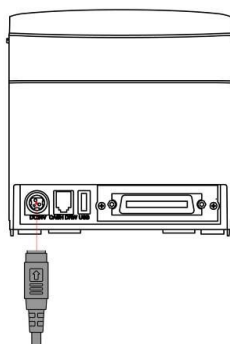


Vertical installation

- 2) Keep the printer far away from water source;
- 3) Do not place the printer in the place exposed to vibration and impact;
- 4) The printer must be safely grounded;
- 5) It is recommended to keep proper space in order to guarantee the reliability and operational convenience of the printer during operation and maintenance.



4.3 Power connection

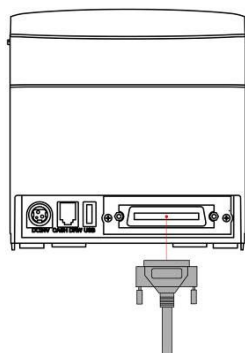


- 1) Ensure that the power switch is turned off;
- 2) Plug the power plug into the power socket on the back of the printer.

⚠ Caution:

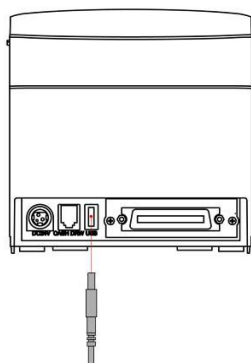
- ✧ When the printer is not in use for a long period of time, disconnect the power cord from the printer.

4.4 Interface connection



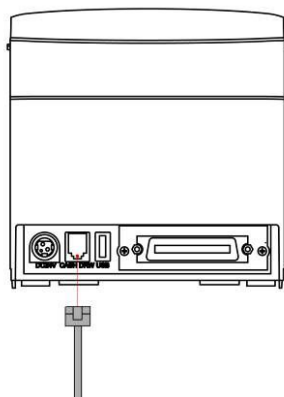
- 1) Ensure the power switch is turned off;
- 2) Insert the interface cable into suitable interface, and fix the cable connector with screw or latch spring;
- 3) Connect the other end of the interface cable to the host.

4.5 USB interface connection



- 1) Ensure that the power switch is turned off;
- 2) As shown in the figure, insert the USB cable into the suitable USB interface, and ensure that it is inserted into place. The other end of USB interface cable should be connected to the host.

4.6 Cash drawer connection



- 1) Ensure that the printer power is turned off;
- 2) Insert the cash drawer cable into the cash drawer interface on the back of the printer.

Caution:

- ✧ Cash drawer interface can only be connected with a cash drawer device (it cannot be connected with a telephone line, etc.).

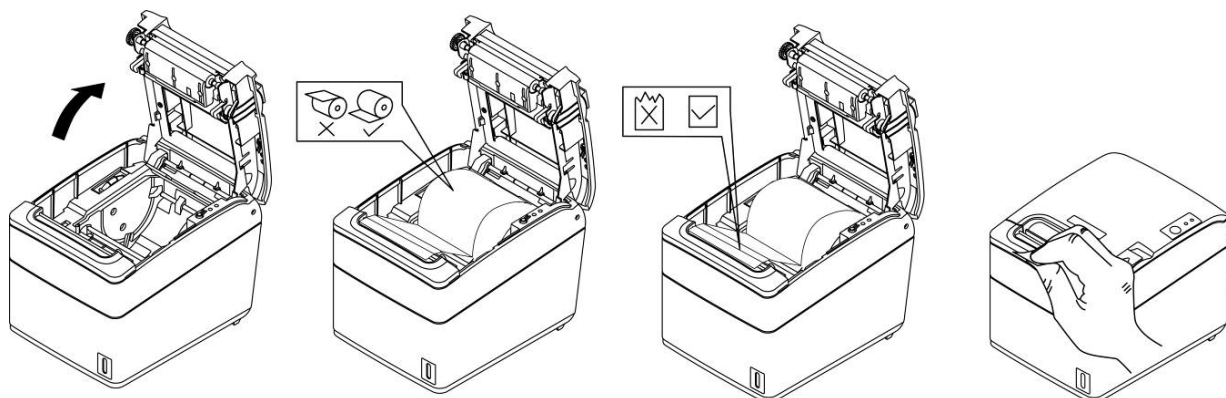
4.7 Paper roll installation

4.7.1 Paper type confirmation

After connecting the power cord and interface cable, confirm paper type after media installation before printing.

4.7.2 Install/replace paper roll

- 1) Turn off the printer;
- 2) Press cover open lever to open the top cover;
- 3) Place the paper roll into the paper cabinet;
- 4) Pull out the paper head of paper roll, close top cover, and tear off the surplus portion of the paper head.



Caution:

- ✧ Adjust the paper guide according to paper specification: adjust the paper guide to maximum width and place paper roll into the paper cabinet; rotate the thumb wheel to adjust the paper guide to suitable place (1mm gap between paper roll and paper guide should be reserved in case the paper roll is jammed); the winding direction of paper roll should meet printer requirements.

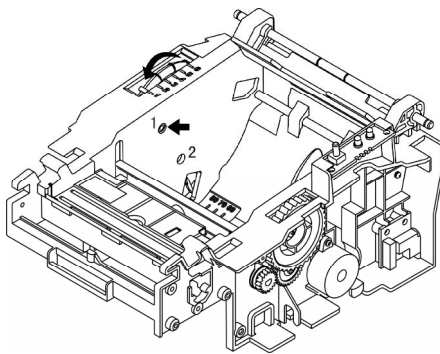
- ✧ Ensure that the paper roll is tightly wound; otherwise paper jam or other failure may occur;
- ✧ Paper roll should be placed stably in the paper cabinet without incline, or the printing will be affected.

4.8 Paper near end adjustment

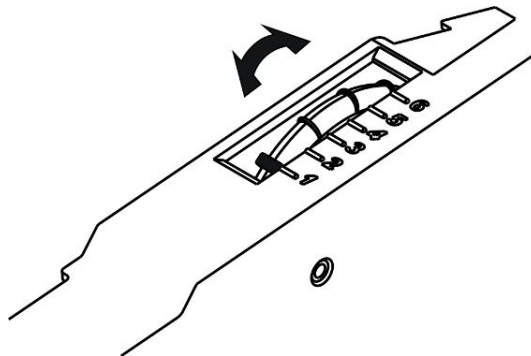
4.8.1 Paper near end position adjustment

RP-600 supports printing in horizontal and vertical positions. Users can realize the detection of remaining paper in horizontal and vertical positions by adjusting the position of paper near end sensor. The default printer setting is horizontal printing and the paper near end sensor is at the bottom of the paper cabinet.

Method of adjusting the paper near end position: First, push the plastic flat spring in position 1 along the direction of the arrow. Then turn the thumb wheel in the direction shown in the figure so that the paper near end sensor starts to rotate. When the plastic flat spring enters position 2, the switching of paper near end sensor position from the horizontal to the vertical is completed.



4.8.2 Adjusting paper remains of paper near end



Amount of remaining paper can be adjusted by adjusting the position of paper near end sensor. Paper near end has six levels with the black mark showing the current level, and the black mark can be adjusted to a different position by turning the thumb wheel so as to change the amount of remaining paper.

4.9 Power-on and self-test

4.9.1 Power-on

- 1) Ensure that the printer is connected to power;
- 1) Turn on the power switch to power on the printer.

4.9.2 Print self-test page

- 1) Ensure that the printer is connected to power, and that paper roll is installed;
- 2) Ensure that the power LED is off and the printer is under power-off condition.
- 3) Press the feed button while turning on the printer power, and then release the button. The printer will print out configuration information and prompt characters "Press and Release FEED key to print characters" and "Press and Hold FEED key to config the printer". Then the printer enters waiting status, and the error LED flashes;
- 4) Pressing the feed button momentarily, the printer will print out a character test sample, and the printing of self-test page is completed. Pressing the feed button for a long time, the printer will enter button configuration mode; for detailed function and operation method of button configuration mode, please refer to Appendix A Button Configuration Table.

4.10 Hexadecimal dumping function

After entering Hexadecimal dumping mode, the printer will print out the data transmitted from the host computer in hexadecimal and their corresponding ASCII characters.

The sample printed under Hexadecimal dumping mode is as follows:

```

Hexadecimal Dump
To terminate hexadecimal dump,
press FEED button three times.

1B 21 00 1B 26 02 40 40 1B 69      . ! . . & . @ @ . i
1B 25 01 1B 63 34 00 1B 30 31      . % . . c 4 . . 0 1
41 42 43 44 45 46 47 48 49 4A      A B C D E F G H I J

*** completed ***

```

Using Hexadecimal dumping mode:

- 1) Entering Hexadecimal dumping mode in the following ways:
 - a. Open the top cover and turn on the printer while pressing the feed button. Release the button after the printer alarms (LED flashes and buzzer beeps).
 - b. Send command "GS (A".
- 2) The printer first prints "Hexadecimal Dump To terminate", and then prints the data transmitted from the host in hexadecimal and their corresponding ASCII characters.
- 3) Exiting from Hexadecimal dumping mode in the following ways:
 - a. Turn off the power, and then restart the printer.
 - b. Press the feed button three times.

Caution:

- ✧ If the hexadecimal data has no corresponding ASCII characters, the printer will print ".";
- ✧ Under Hexadecimal dumping mode, only commands DLE EOT, DLE ENQ, or DLE DC4 are valid.
- ✧ The data of the last character line can be printed by pressing down the feed button.

5 Routine maintenance

Caution:

- ✧ Before starting routine maintenance, ensure that the printer power is turned off.
- ✧ Do not use organic solvents like gasoline or acetone.
- ✧ When cleaning sensors, do not turn on the printer power until the pure alcohol has completely evaporated.
- ✧ It is recommended that the maintenance cycle should not be longer than one month.

5.1 Cleaning print head and platen roller

Follow the steps below to clean the print head and platen roller:

- 1) Turn off the printer and open the top cover;
- 2) If the printing was just finished, please wait for the print head to cool down completely;
- 3) Wipe off the dust and stains on the surface of print head and platen roller with alcohol cotton (it should be wrung out);
- 4) After the alcohol is completely evaporated, close the top cover.

5.2 Cleaning mark sensor

When the printer cannot identify the mark effectively, the mark sensor should be cleaned.

The cleaning steps are as follows:

- 1) Turn off the printer;
- 2) Press the cover open lever to open the top cover;
- 3) Take off the transparent sensor cover;
- 4) Wipe off the dust and stains on the surface of the sensor with soft cotton cloth dipped with pure alcohol (it should be wrung out);
- 5) Install the sensor cover after the pure alcohol is completely evaporated, then close the top cover and finish sensor cleaning.

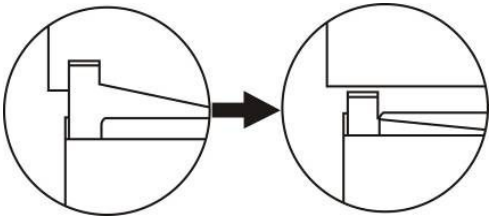
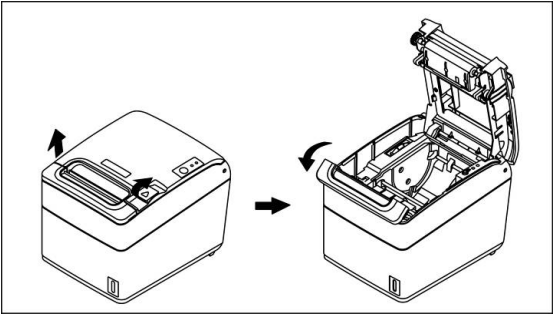
5.3 Clearing paper jam

Steps for clearing paper jam are as follows:

- 1) Turn off the printer power and open the top cover.
- 2) Clear paper jam, and then close the top cover.
- 3) Turn on the printer power again to reset the cutter.

Caution:

- ✧ If the cutter is jammed and the top cover can not be opened, please put your hands at both sides of the cutter cover and turn the cutter cover forward, and then the top cover can be opened.
- ✧ After clearing the paper jam, do not restart the printer until the top cover is closed and the cutter cover is reset.



Before recovery

After recovery

6 Interface signal

6.1 Parallel interface

Parallel interface can work in IEEE 1284 compatible mode or nibble mode. The interface is 36PIN socket.

The Interface is defined as below:

Pin No.	Signal source	Signal definition
1	H	nStrobe
2	H	Data 0 (Least Significant Bit)
3	H	Data 1
4	H	Data 2
5	H	Data 3
6	H	Data 4
7	H	Data 5
8	H	Data 6
9	H	Data 7 (Most Significant Bit)
10	P	nAck
11	P	Busy
12	P	Perror
13	P	Select
14	H	nAutoFd
15		Not defined
16		Logic Gnd
17		Chassis Gnd
18	P	Peripheral Logic High
19		Signal Ground (nStrobe)
20		Signal Ground (Data 1)
21		Signal Ground (Data 2)
22		Signal Ground (Data 3)
23		Signal Ground (Data 4)
24		Signal Ground (Data 5)
25		Signal Ground (Data 6)
26		Signal Ground (Data 7)
27		Signal Ground (Data 8)
28		Signal Ground (PError, Select, and nAck)
29		Signal Ground (Busy and nFault)
30		Signal Ground (nAutoFd, nSelctIn, and nInit)
31	H	nInit
32	P	nFault
33		Not defined
34		Not defined
35		Not defined
36	H	nSelectIn

6.2 Serial interface

The serial interface of the printer is compatible with RS-232 standard, and its interface socket is DB-25.

PIN No.	Signal definition
PIN1	Frame ground
PIN2	TXD
PIN 3	RXD
PIN 4	RTS
PIN 5	Not connected
PIN6	DSR
PIN 7	Signal ground
PIN 8~19	Not connected
PIN 20	DTR
PIN 21~25	Not connected

User can query interface setting status via printing self-test page. The default settings of serial interface are as follows:

Baud rate: 115200bps; Data bits: 8; Parity: none; Stop bit: 1; Handshake: DTR/DSR.

6.3 USB interface

1) Parameters

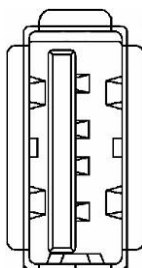
Data transmission: Support USB2.0 protocol.

Connector (Printer end): USB A type socket. Support USB HUB.

2) Interface signal definition and function

Pin No.	Signal name	Description
1	VBUS	+5V
2	DATA-	Data-
3	DATA+	Data+
4	GND	Ground

3) Interface connector



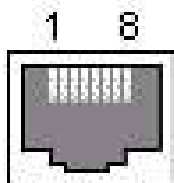
6.4 Ethernet interface

1) Interface features

- Support 10/100MBASE-T communication standard
- Compatible with Ethernet II standard frame type
- LED shows network connection status and data transmission status

- Support 9100 port print
- Support status back
- Support parameter configuration
- Support firmware on-line update
- Support printer status query and interface module maintenance based on HTTP

2) Interface signal definition



Interface adopts 10BASE-T standard which complies with IEEE802.3. The interface signal is defined as below:

Pin No.	Signal name	Description
1	TX+	Data transmission +
2	TX-	Data transmission -
3	RX+	Data receiving +
4	NC	Not connected
5	NC	Not connected
6	RX-	Data receiving -
7	NC	Not connected
8	NC	Not connected

Interface module pin list

6.5 WLAN interface

1) Interface features

- Support 802.11b, 802.11g communication
- Support 9100 port printing, LRP printing
- Support status back
- Support parameter configuration
- Support firmware on-line update
- Support HTTP function

2) Protocols supported

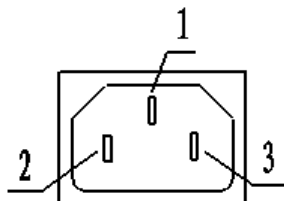
- IP
- ARP
- ICMP
- TCP
- UDP
- DHCP
- TFTP

➤ HTTP

Wireless interface uses wireless USB network card. For the main specification, please consult the local dealer or manufacturer.

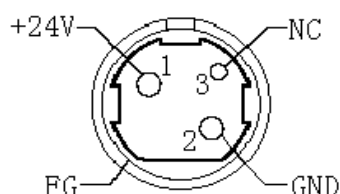
6.6 Power interface definition

Power interface signal definition:



Pin No.	Signal name
1	E
2	L
3	N

24V power interface definition



6.7 Cash drawer interface signal definition

1) Electrical features

- Driving voltage: DC 24 V
- Driving current: Max. 1 A
- Cash drawer status detection signal: "L" = 0~0.5 V "H" = 3.3 V

2) Cash drawer interface socket uses RJ-11 6P connector:



3) Interface signal definition

Pin No.	Signal	Functions
1	FG	Frame Ground
2	DRAWER 1	Cash drawer 1 driving signal
3	DRSW	Cash drawer status detection signal
4	VDR	Cash drawer driving power
5	DRAWER 2	Cash drawer 2 driving signal
6	GND	Circuit share ground

 Caution:

- ✧ Do not connect or disconnect communication cable plug when the printer is powered on;
- ✧ Communication cable should be far away from strong current;
- ✧ Communication cable should adopt shielded cable.

7 Troubleshooting

When the printer has any problem, refer to this chapter for solution.

If the problem still cannot be solved, please contact your local dealer or manufacturer for assistance.

7.1 Printer doesn't work

Problem	Possible causes	Solution
LED is off and the printer doesn't work	Printer is not connected to power supply	Connect the printer to power supply
	Printer power is off	Turn on the printer power
	Circuit board is damaged	Contact your local dealer or manufacturer

7.2 Error LED and Buzzer

Problem	Possible causes	Solution
Error LED flashes or buzzer beeps	Paper end	Replace roll paper
	Cutter error	Refer to 5.3 to clear cutter error.
	Top cover lift-up	Close top cover
	Print head is overheating	Turn off printer power and wait for the print head to cool down
Error LED is always on and buzzer keeps beeping	The printer has serious malfunction	Contact your local dealer or the manufacturer

7.3 Problems during printing

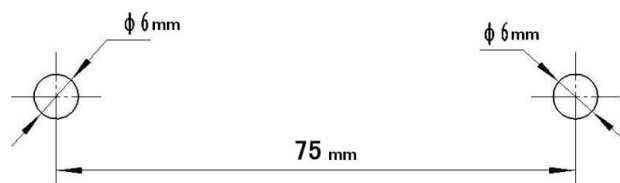
Problem	Possible causes	Solution
Paper cannot be sent out normally	Paper jam	Open top cover, check paper path and cutter, and clear paper jam
Printer starts printing but stops suddenly during printing	Paper jam	Open top cover, check the cutter, and clear paper jam
Paper is not cut off	Paper jam	Open top cover, check the cutter, and clear paper jam
Printout is not clear or has stains	Paper roll is not installed correctly	Make sure that the paper roll has been installed correctly
	Paper is out of specification	Use recommended thermal paper
	Dirty print head or platen roller	Clean the print head or the platen roller
	Print darkness is too low	Increase the print darkness as needed
Vertical print content is missing	Dirty print head or platen roller	Clean the print head or the platen roller
	Print head error	Contact your local dealer or manufacturer

8 Options

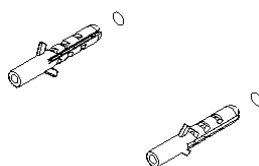
➤ Wall mounting module

The printer can be mounted on the wall according to the instructions below:

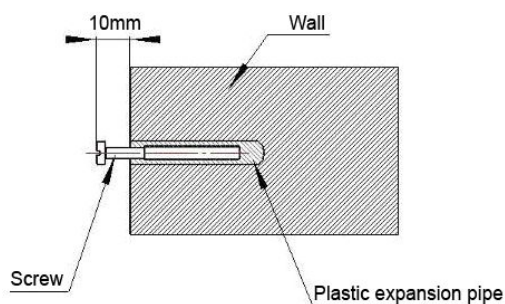
- 1) Drill two holes (diameter: 6mm; depth: 50mm; space between two holes: 75mm) on the wall.



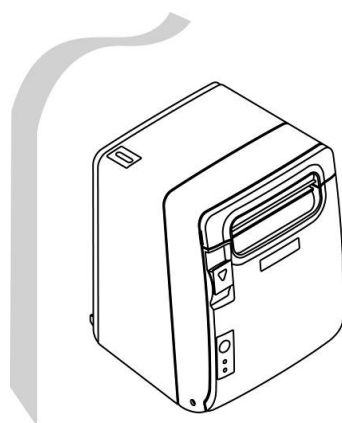
- 2) Insert the plastic expansion pipes into the holes and make sure that the end of expansion pipe is parallel to the wall surface.



- 3) Screw the screw into the plastic expansion pipe and keep 10mm space between the screw head and the wall surface.



- 4) Adjust the paper near end sensor position (please refer to 4.8.1) and mount the printer as shown in the following figure.



9 Power management

The power management of RP-600 has four operation modes: Off, standby, Active, Sleep.

- 1) Printer will enter standby mode after powering on the printer or completing printing task.
- 2) Printer will enter sleep mode if there is no printing task for 2 minutes in standby mode.
- 3) Printer will awake automatically and enter active mode when a printing task comes, and will enter standby mode again after completing the printing task.

Appendix A: button configuration table

PARAMETER SETTING BY FEED BUTTON									
MAIN MENU									
Exit	->1								
Print Self Test	->2								
Configuration	->3	CONFIGURATION							
		Exit Without Save	->1						
		Exit With Save	->2						
		Communication	->3	Back To Last Menu	->1				
				USB Interface	->2	USB MODE:API MODE			
						Back To Last Menu	->1		
						WinDriver Mode	->2		
						API Mode	->3		
				Serial Interface	->3	SERIAL INTERFACE			
						Back To Last Menu	->1		
						Baud Rates	->2	BAUD RATES:19200bps	
								Back To Last Menu	->1
								9600bps	->2
								19200bps	->3
								38400bps	->4
								57600bps	->5
								4800bps	->6
								2400bps	->7
								1200bps	->8
								115200bps	->9
						Parity	->3	PARITY:NONE	
								Back To Last Menu	->1
								None	->2
								Odd	->3
								Even	->4
						Data Bits	->4	DATA BITS : 8 Bits	
								Back To Last Menu	->1
								7 Bits	->2
								8 Bits	->3
						Stop Bit(s)	->5	STOP BITS: 1 Bit	
								Back To Last Menu	->1
								1 Bit	->2
								2 Bits	->3
						Handshaking	->6	HANDSHAKING:DTR/DSR	
								Back To Last Menu	->1
								DTR/DSR	->2
								XON/XOFF	->3
						Data Receive Error	->7	DATA ERROR SETTING:Ignored	
								Back To Last Menu	->1

				Ignored	->2
				Print '?'	->3
		Rx Buff Size	->4	RX BUFFER SIZE:4K Bytes	
				Back To Last Menu	->1
				4k Bytes	->2
				45 Bytes	->3
				64K Bytes	->4
	Mechanism & Hardware	->4	HARDWARE SETTINGS		
			Back To Last Menu	->1	
		Mark Sensor	->2	MARK SENSOR: Disable	
				Back To Last Menu	->1
				Enable	->2
				Disable	->3
		Cutter	->3	CUTTER:	
				Back To Last Menu	->1
				Cut Mode Settings	->2
				CUT MODE SETTINGS: Default CutMode	
				Back To Last Menu	->1
				Enable	->2
				Disable	->3
				Full Cut Mode	->4
				Partial Cut Mode	->5
				Default Cut Mode	->6
				Auto Cut Settings	->3
				AUTO CUT SETTINGS: No use this function	
				Back To Last Menu	->1
				Cut paper when cover is closed	->2
				No cut paper when cover is closed	->3
				Cut paper when power on	->4
				No cut paper when power on	->5
				Disable	->6
		Buzzer	->4	BUZZER: Normal Volume	
				Back To Last Menu	->1
				Low Volume	->2
				Normal Volume	->3
				High Volume	->4
				Higher Volume	->5
				Highest Volume	->6
				Disabled	->7

		Power Supply	->5	POWER SUPPLY: Normal					
				Back To Last Menu	->1				
				Normal	->2				
				Low Power Mode	->3				
		Print Settings	->5	PRINT SETTINGS					
				Back To Last Menu	->1				
				Darkness Settings	->2				DARKNESS SETTING: Normal
					Back To Last Menu				->1
					Low				->2
					Normal				->3
	High				->4				
			Extra High	->5					
	Paper Roll Width	->3	PAPER ROLL WIDTH:80.0mm						
			Back To Last Menu	->1					
			54~59mm	->2			Select A Width		
				Back To Last Menu			->1		
				54mm			->2		
				55mm			->3		
				56mm			->4		
				57mm			->5		
				58mm			->6		
			59mm	->7					
			60~65mm	->3	Select A Width				
				Back To Last Menu	->1				
				60mm	->2				
				61mm	->3				
				62mm	->4				
				63mm	->5				
64mm				->6					
65mm			->7						
66~71mm	->4	Select A Width							
	Back To Last Menu	->1							
	66mm	->2							
	67mm	->3							
	68mm	->4							
	69mm	->5							
	70mm	->6							
71mm	-> 7								
72~77mm	->5	Select A Width							
	Back To Last Menu	->1							
	72mm	->2							
	73mm	->3							
	74mm	->4							
	75mm	->5							
	76mm	->6							
77mm									

			78~82mm		->6	Select A Width		
					Back To Last Menu		->1	
					78mm		->2	
					79mm		->3	
					80mm		->4	
					81mm		->5	
					82mm		->6	
			Left Margin		->4	LEFT MARGIN:7mm		
					Back To Last Menu		->1	
					0mm		->2	
		1mm			->3			
		3mm			->4			
		5mm			->5			
		7mm			->6			
		9mm			->7			
		Right Margin		->5	RIGHT MARGIN:9mm			
				Back To Last Menu		->1		
				0mm		->2		
				1mm		->3		
				3mm		->4		
				5mm		->5		
				7mm		->6		
				9mm		->7		
		CR Command		->6	CR COMMAND: Disable			
				Back To Last Menu		->1		
				Enable		->2		
				Disable		->3		
		Code Page		->7	CODE PAGE SETTING			
				Back To Last Menu		->1		
				Print all codepages		->2		
				Select a codepage		->3		
		Save Paper Level		->8	SAVE PAPER LEVEL			
				Back To Last Menu		->1		
				Disable		->2		
				25%		->3		
				50%		->4		
				75%		->5		
				100%		->6		
		Two-color Verify		->9				
		Gray Bmp Verify		->10				
	Paper Sensor Settings		->6	PAPER NEAR END SETTINGS				
			Back To Last Menu		->1			
			Paper Low Alarm		->2			
				PAPER LOW ALARM: Enable				

			Back To Last Menu	->1			
			Enable	->2			
			Disable	->3			
		Stop Print When PAPER Low	->3	STOP PRINT WHEN PAPER LOW: Disable			
			Back To Last Menu	->1			
			Enable	->2			
			Disable	->3			
		Paper Near End Sensor	->4	PAPER NEAR END SENSER: Enabled			
			Back To Last Menu	->1			
			Enable	->2			
			Disable	->3			
		Set Default Config	->7	SET DEFAULT CONFIGURATION			
			Back To Last Menu	->1			
			Set Printer To Default Configuration	->2			
		FONTA/FONTB Settings	->8	Current Font:FONTA			
		Back To Last Menu	->1				
		Select FONTA	->2				
		Select FONTB	->3				
		Select UDFONTA	->4				
		Select UDFONTB	->5				
	Beep settings	->9	Beep settings:Disabled				
		Back To Last Menu	->1				
		Enable External Herald	->2				
		Enable Internal buzzer	->3	BEEP MODE			
			Back To Last Menu	->1			
			Mode 1	->2			
			Mode 2	->3			
			Mode 3	->4			
			Mode 4	->5			
	Mode 5	->6					
	All Beep disabled	->4					
	Enter code,then hold Button Down at least 1 second to validate						
Sensor Test	->4	Sensor Test Mode: ERROR LED state will change according to sensor					

		state To EXIT, hold button down at least 1 second				
Cutter Test	->5					
Print Statistics	->6	BTP-880NP STATISTICS				
		TCUT	:0			
		TLFS	:0			
		ONTIME	:0			
Calibration	->7					
E05 Configuration	->8	E04CONFIGURATION				
		Exit Without Save	->1			
		Exit With Save	->2	Saving config, please wait. Configurations have been saved, please repower the printer		
		Reset JK-E05 Config	->3			
		Print Settings	->4	IP Address: MAC Address: SUBNET Mask: GATEWAY: Print Port:		